Class 23: Confinement and Quantization: Part 2

In the last class we have:

1) Examined the potential experienced by an electron as a function of position in a solid.
2) Developed a reasonable approximation of the same, such that curves become square shaped potential wells.
3) Recognized the idea of confinement of an electron. In particular, noted that some electrons are confined to the extent of the solid which could be of the order of a few meters, and are called the nearly free electrons. Some electrons are confined to the vicinity of individual ionic cores and are called bound electrons. Only if the electron has escaped the solid it becomes a truly free electron.
4) Examined the analogy of waves on a string to understand the effect of confinement on waves.
5) Noted that confinement leads to quantization of energy.

When an electron is trapped in a potential well, its wave like behavior interacts with the confinement it faces and responds by adopting only specific wavelengths consistent with the extent of confinement. These wavelengths correspond to fixed values of energy, resulting in quantization of the energy of the confined electron.

Confinement within an extent 'a', places the restriction on wavelengths that electron can adopt such that

$$\frac{n\lambda}{2} = a$$

Where n has to be an integer. This implies that the allowed values of k are such that

$$k = \frac{2\pi}{\lambda} = \frac{n\pi}{a}$$

Since allowed energy values are related to allowed k values as below,

$$E = \frac{\hbar^2 k^2}{2m}$$

The only values of energy permitted to a confined electron are:

$$E = \frac{\hbar^2 n^2 \pi^2}{2ma^2}$$

We arrived at the above result using the analogy of waves on a string. Let us now examine the problem of a confined electron using quantum mechanical principles. To use quantum mechanics we have to obtain the wave function corresponding to the system. To obtain the wave function,

we need to solve the Schrödinger wave equation with the conditions placed on the system. The wave function we will obtain will contain the properties of the system. The Schrödinger wave equation in one dimension is as follows:

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+V(x)\right)\psi(x)=E\psi(x)$$

A free electron does not experience any potential, therefore $V(x)=0$

Therefore, for a free electron,

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\right)\psi(x)=E\psi(x)$$

Rearranging, we have:

$$\frac{d^2\psi(x)}{dx^2}=\frac{-2mE}{\hbar^2}\psi(x)$$

In other words, a function of x, that is differentiated twice, results in a constant times the original function. Simply by observation, we can conclude that trigonometric functions of the type

$$\psi(x)=Acos(kx)+Bsin(kx)$$

Will solve the differential equation above, where we can identify k with the wave vector we have discussed earlier – it has the correct dimensions since kx should be dimensionless.

Substituting for $\psi(x)$ in the left hand side of the equation above, we have:

$$\frac{d^2\psi(x)}{dx^2}=\ -k^2[Acos(kx)+Bsin(kx)]$$

Therefore:

$$k^2=\frac{2mE}{\hbar^2}\Rightarrow E=\frac{\hbar^2k^2}{2m}$$

The k, being a one dimensional wave vector, can adopt positive as well as negative values. Since the above analysis does not place any restrictions on the value of k, there are no restrictions on the values that E can adopt. This result is consistent with the results obtained through the analogy of waves on a string that is tied on one end only.

Since E is proportional to k^2, and the other quantities in the equation are constants, the plot of E vs k results in a parabola as shown in Figure 23.1 below.

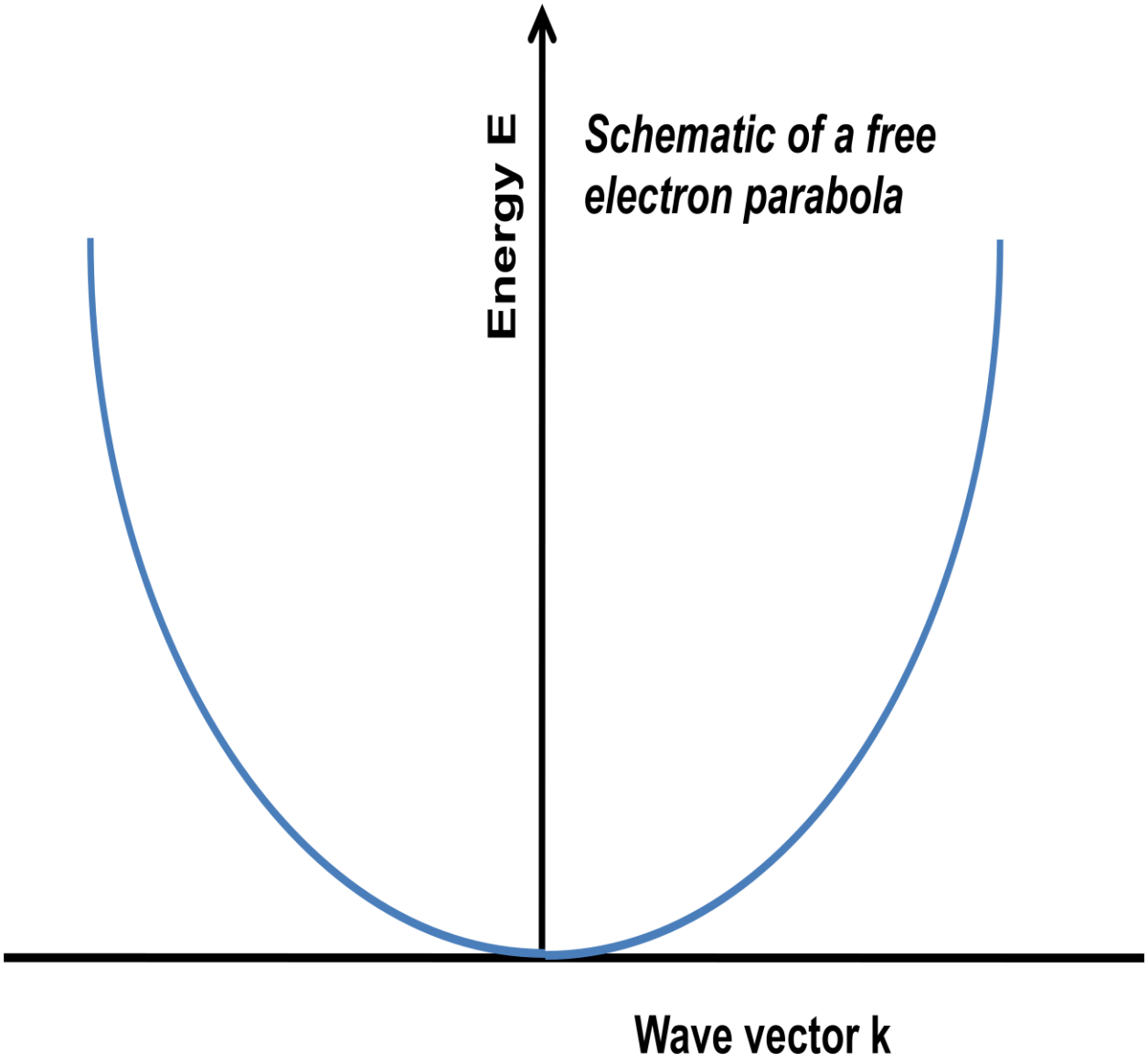


Figure 23.1: Plot of E vs k for a free electron. All values of k, and hence all values of E, are permitted. The parabola is continuous, and E can adopt any value along the y-axis.

Let us now consider the case of an electron trapped in a potential well. Figure 23.2 below shows an electron trapped in a potential well.

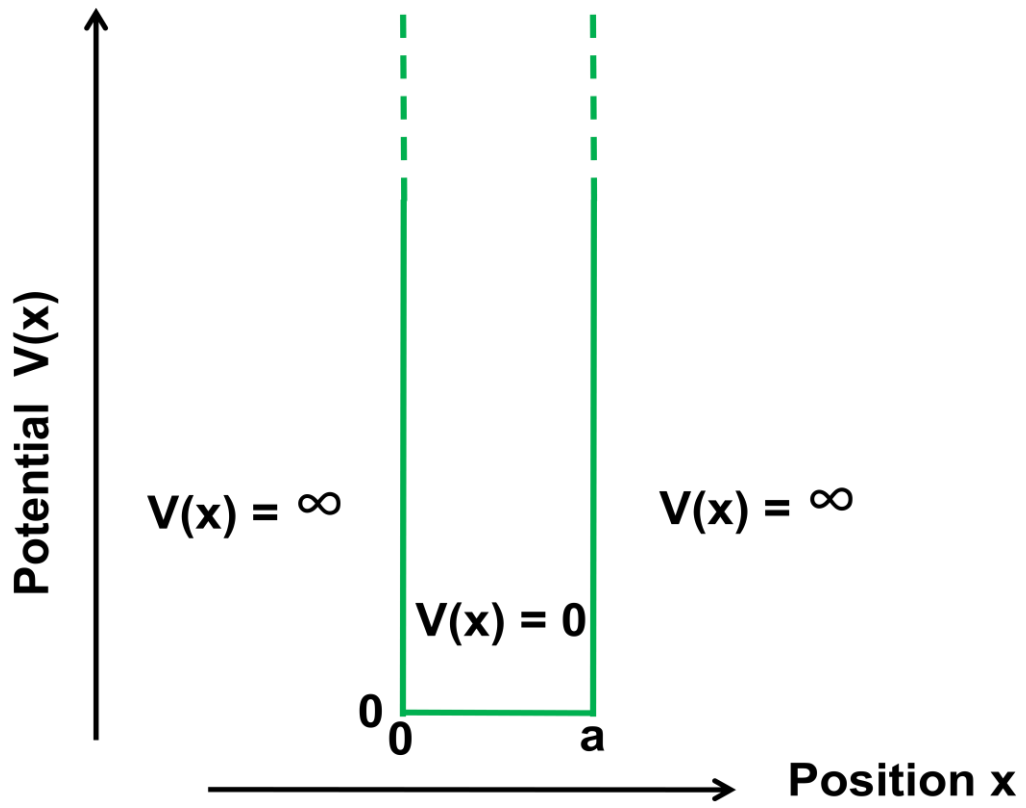


Figure 23.2: A potential well in which an electron is trapped. The potential is zero between the walls of the potential well, and becomes infinity at and beyond the walls.

Within the well $V(x) = 0$, therefore the form of the solution is the same as before.

$$\psi(x) = Acos(kx) + Bsin(kx)$$

While this looks similar in form to what we obtained earlier for a free electron, it differs in one important detail. Due to the infinite potentials at the walls, the probability of finding the electron outside the walls is zero. This means the wave function is zero outside the walls. To avoid a discontinuity at the walls, the wave function inside the potential well must also drop to zero at the walls. Therefore, $\psi(0) = 0$, and $\psi(a) = 0$.

Since

$$\psi(0) = A$$

Therefore $A = 0$

Which leads us to

$$\psi(a) = Bsin(ka) = 0$$

This is possible when

$$ka = n\pi$$

Therefore, for an electron trapped in a potential well, the only values of wave vector k, that are permitted are:

$$k = \frac{n\pi}{a}$$

a result that is identical to the one we obtained when we examined a string tied at both ends.

Since k can only have specific values, only corresponding values of energy are permitted. In other words, energy of the electron is now quantized. Since energy is related to the k vector through the same relationship as before,

$$E = \frac{\hbar^2 k^2}{2m}$$

the plot of the E vs k is still a parabola. It is just that not all points on the parabola are permitted. Only specific values of the k vector are permitted on the x-axis, and hence only the corresponding values of E are permitted on the y-axis. The allowed values are shown in the schematic in Figure 23.3 below.

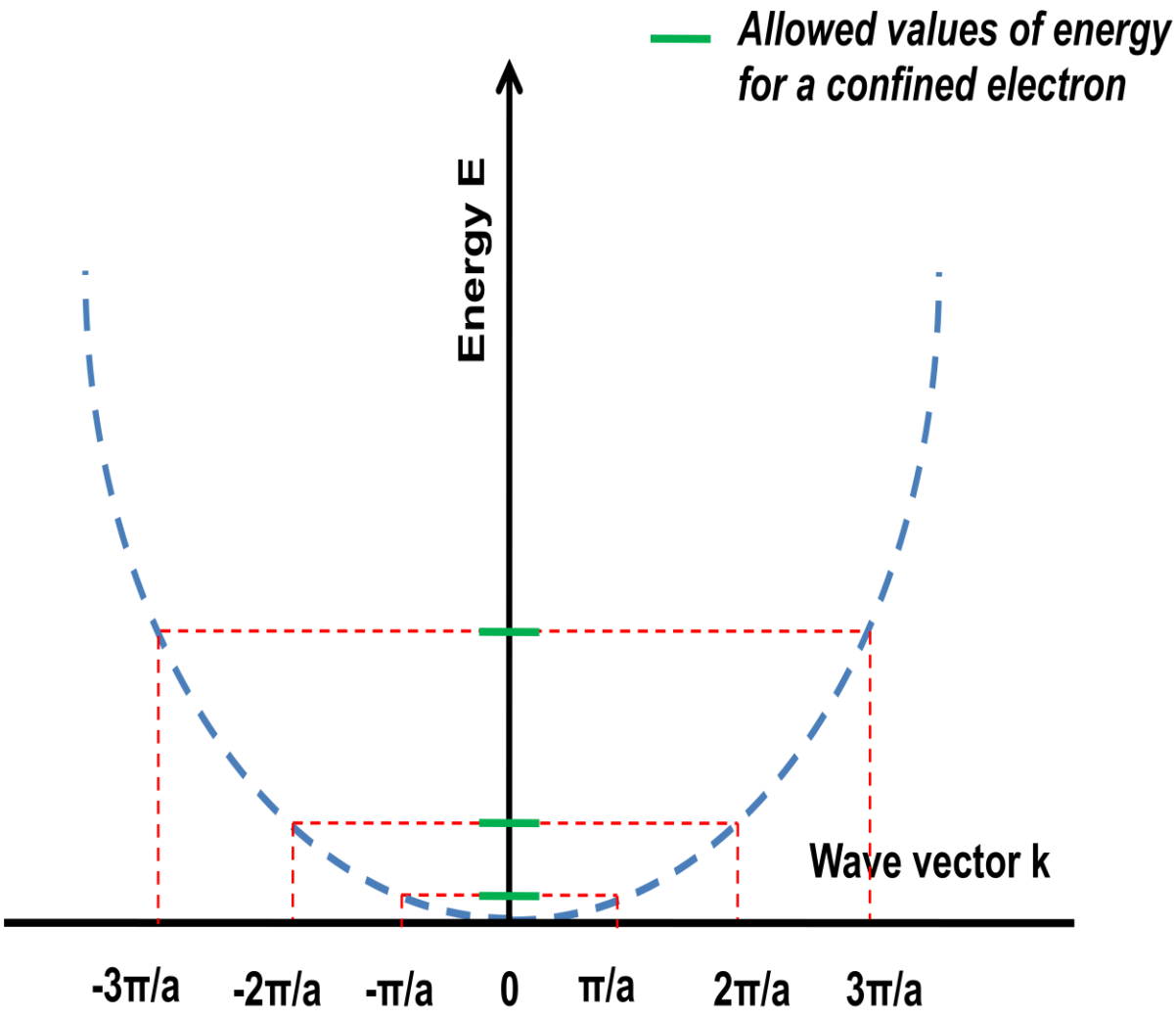


Figure 23.3: Schematic of the allowed values of k vector, and the corresponding values of E, for an electron trapped in a potential well.

There is one additional point to note, which is regarding the level of confinement faced by the different electrons in the solid. In the equation for the wave vector:

$$k = \frac{n\pi}{a}$$

'a' represents the extent of confinement. The extent of confinement of nearly free electrons, and bound electrons, is shown in Figure 23.4 below.

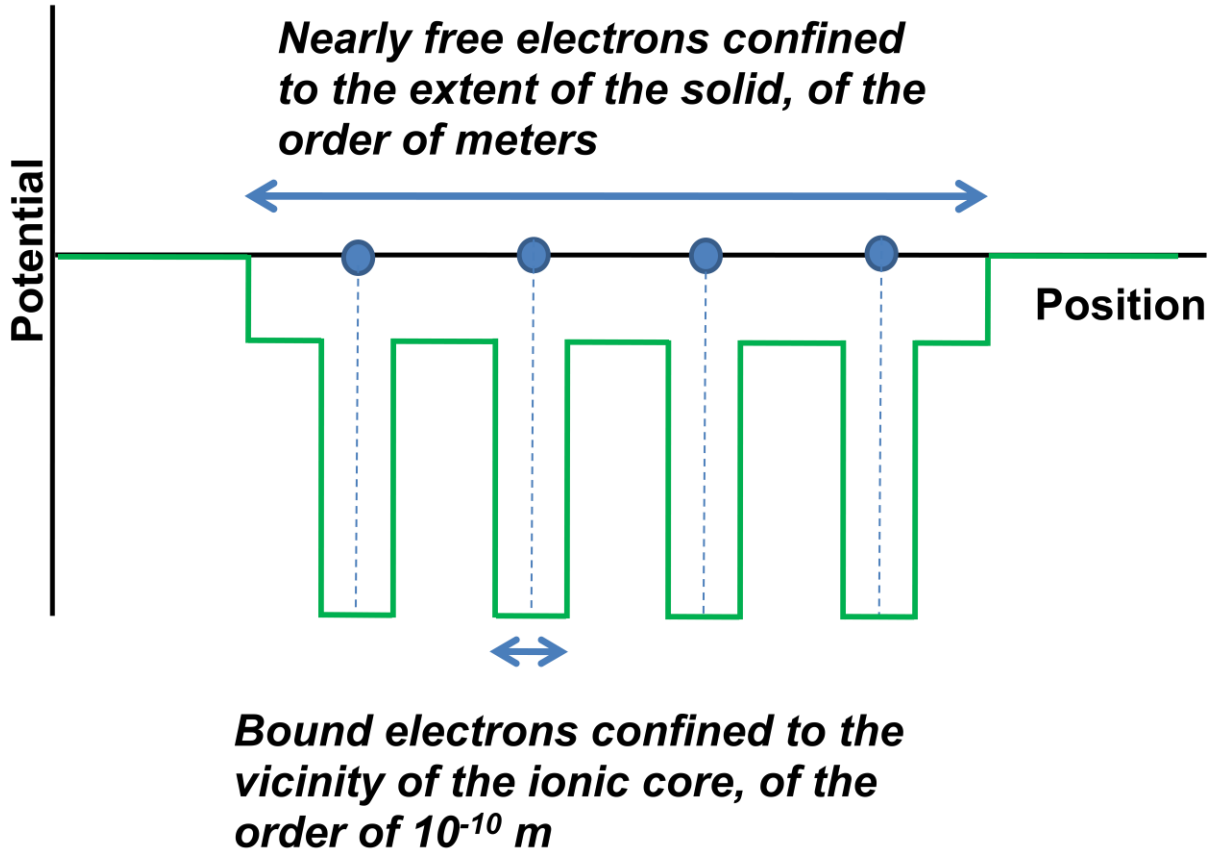


Figure 23.4: The difference in the extent of confinement of nearly free electrons, and bound electrons.

For nearly free electrons, the extent of confinement is the extent of the solid, and is in the order of meters. Therefore the value of 'a' is relatively large in this case and since it is in the denominator, it causes the adjacent values of k to be closely spaced. This in turn causes the allowed values of energy to also be relatively closely spaced as well.

For bound electrons, the extent of confinement is the length scale of an ionic core, and is in the order of one Å or 10^{-10}m. Therefore the value of 'a' is extremely small in this case and since it is in the denominator, it causes the adjacent values of k to be spaced extremely widely apart. This in turn causes the allowed values of energy to also be relatively widely spaced as well.

In summary we find that we are able to conclude that confinement leads to quantization and that the more narrow the region of confinement, wider spaced are the corresponding allowed energy values. In the next class we will examine the consequences of quantization further.